

APT-BENCH: PATIENT-DISJOINT CELL TYPING FROM SINGLE-CELL APTAMER PROFILES

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ABSTRACT

Evaluation of single-cell aptamer cell typing must distinguish additional cells from additional patients. We introduce APT-Bench, a patient-disjoint benchmark specification and reference evaluation comprising 361,792 cells, 293 aptamer measurements, and paired-transcriptomic annotations for five lineages and 27 subtypes from 40 participants. Five fixed outer patient folds and subject-balanced pooled Macro-F1 define the new-patient evaluation contract. In the available XGBoost and Soft-cascade Transformer controls, cell-level splitting produces higher coarse and fine Macro-F1 in all four comparisons, with increases of 0.050–0.108 on the historical cell-weighted metric. These protocol contrasts show that held-out-cell performance can give an optimistic account of new-patient generalization, without isolating a causal leakage effect. Classical and Transformer references establish performance under the patient-disjoint contract; Soft-cascade is a hierarchy-aware baseline rather than a claimed algorithmic contribution. Paired RNA and patient aggregation provide secondary context, not independent clinical validation. APT-Bench centers the statistical unit of evaluation in this measurement modality, while complete data access and portable reproduction remain explicit release requirements.

1 INTRODUCTION

Single-cell aptamer profiling creates a targeted molecular representation of blood-derived immune cells. To assess whether this representation supports cell typing, a model must be evaluated on patients absent from training, not merely on additional cells from familiar patients. The distinction matters because a large cell count does not create an equally large set of independent biological samples.

APT-Bench addresses this evaluation problem with 361,792 cells from 40 participants, 293 aptamer measurements per cell, and paired-transcriptomic annotations defining five lineages and 27 subtypes. Its central contribution is a patient-disjoint cell-typing benchmark specification and empirical evaluation for the APT modality. We fix the outer patient folds, state the label ontology and input contract, and report subject-balanced pooled Macro-F1 so that highly sampled patients do not dominate the score.

The central empirical finding concerns the evaluation protocol. In the available XGBoost and Soft-cascade comparisons, cell-level splitting gives higher coarse and fine Macro-F1 than patient-disjoint evaluation in all four model-resolution comparisons. The increases range from 0.050 to 0.108 on the historical cell-weighted metric. Thus evaluation on held-out cells from known patients can give a substantially more favorable picture than evaluation on new patients. These protocol comparisons are not a matched causal estimate of leakage alone; they nevertheless show why the intended generalization unit must be fixed before interpreting performance.

Classical and Transformer models provide reference performance under that contract. Soft-cascade Transformer is one hierarchy-aware reference, not a separate algorithmic contribution or the basis for a superiority claim. We retain an internal paired-RNA reconstruction diagnostic and a short patient-level application study to contextualize the benchmark, rather than treating them as additional headline contributions. Subject-cell sampling experiments are preserved in a separate supporting report.

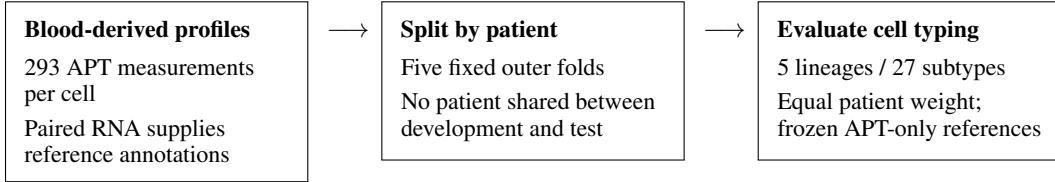


Figure 1: The primary APT-Bench contract. Patient-disjoint outer folds evaluate cell typing in new individuals; subject-normalized confusion pooling assigns equal total scoring weight to each patient. Cell-random splitting is a separate diagnostic, not a substitute for this generalization target.

The paper makes one contribution with three concrete components:

- **A specified evaluation contract:** APT-only lineage and subtype prediction with a fixed ontology, patient-disjoint folds, and explicit cell-weighted versus subject-balanced metrics.
- **Evidence that the split changes the conclusion:** the evaluated cell-level split controls consistently overstate performance relative to the new-patient protocol, motivating patient-disjoint reporting.
- **Reference results and an auditable interface:** classical and neural predictions, metric code, split manifests, and documented artifact checks establish comparison points for subsequent APT models.

A complete public benchmark also requires accessible data, verified permissions, and portable reproduction. Those release requirements remain explicit where unfinished; a public source repository alone does not establish their completion. The study does not claim validated blood-based diagnosis, a universal sampling law, or priority over all prior aptamer evaluation resources.

2 RELATED WORK

Blood-based characterization. Multi-cancer detection commonly uses circulating DNA, methylation, fragmentomics, or bulk proteins (Cohen et al., 2018; Zeng et al., 2026; Wan et al., 2025; Kahwati et al., 2025). APT-Bench addresses the complementary evaluation question of identifying immune cells from single-cell aptamer measurements in new patients. Its disease-associated case study is secondary; within-cohort classification is not prospective screening.

Single-cell representations and hierarchy. Geneformer, scGPT, and scFoundation motivate large transcriptomic representations, while controlled studies show that simple baselines can remain competitive (Theodoris et al., 2023; Cui et al., 2024; Hao et al., 2024; Kedzierska et al., 2025). Cell annotation also motivates hierarchical losses and architectures (Jia et al., 2025; Cultrera di Montesano et al., 2026). We use flat, HCE, and soft-cascade Transformers alongside classical models as APT references. Because their complete objectives are not matched, this ladder illustrates the evaluation setting rather than constituting a new learning algorithm or component ablation.

Evaluation across cells and subjects. Reusable single-cell benchmarks separate datasets, models, and metrics and adopt sample- or study-disjoint evaluation (Luecken et al., 2025; Liu et al., 2025; Shitov et al., 2025; Liu et al., 2026). APT-Bench makes the statistical unit explicit for APT cell typing and contrasts new-patient evaluation with cell-level split controls. Gradient-boosted trees provide strong tabular references (Grinsztajn et al., 2022; Erickson et al., 2026). Separate sampling and feature-budget reports do not constitute additional primary contributions or resolve biological versus technical confounding.

3 BENCHMARK SETTING

3.1 BLOOD PROFILES AND REFERENCE LABELS

APT-Bench is derived from peripheral-blood specimens from 40 participants in five cancer groups (CRC, GC, HCC, PC, and BTC) and healthy controls, with 6–8 participants per group. The paired

assay measures transcriptomic state and 293 target-specific nucleic-acid aptamers on the same cells. Transcriptomic counts were processed with Seurat (Hao et al., 2021); predicted doublets and low-quality cells were removed, and clusters were manually annotated using canonical markers from prior PBMC and pan-cancer studies (McGinnis et al., 2019; Zhang et al., 2024; Yang et al., 2024). These annotations define five lineages and 27 subtypes. They are operational reference labels, not independently measured biological truth.

The linked post-QC cohort contains 374,854 cells. Excluding the uncertain *Unknown* subtype leaves 361,792 benchmark cells. Each subtype has one fixed parent lineage. Disease is a six-class patient attribute, not the root of the cell hierarchy: any given lineage can occur in multiple disease groups. The benchmark therefore connects cell and patient tasks through aggregation, rather than imposing a disease-to-lineage-to-subtype label tree.

The construction checklist and unresolved acquisition and annotation fields are recorded in Appendix C.1. In particular, the analyzed data do not establish a validated one-drop input-volume requirement. The scope here is characterization from the profiled blood cells, not end-to-end validation of a small-volume diagnostic assay.

3.2 PATIENT-DISJOINT CELL-TYPING CONTRACT

The evaluation has a primary task and a secondary application. **Cell identity** predicts lineage and subtype from APT alone. **Patient characterization** predicts disease-group membership from patient summaries of measurements or cell-model outputs. The latter is an exploratory association task: with one acquisition per participant and incomplete acquisition metadata, disease biology and technical structure cannot be identified separately. Open-set subtype discovery is outside the contract; all primary cell tasks use the fixed five-lineage/27-subtype ontology.

Five frozen disease-stratified folds each hold out eight patients. All cells from a patient remain together; preprocessing, model selection, and fitting are restricted to development patients. We pool predictions across the five out-of-fold (OOF) partitions. A cell-random split is retained only as a leakage diagnostic, not as evidence of generalization to new patients. The release interface records cell and patient IDs, ontology, folds, predictions, and metrics (Appendix C.4). Full data access and a portable end-to-end evaluator remain release requirements.

3.3 METRICS AND STATISTICAL UNITS

Coarse and fine subject-balanced pooled Macro-F1 (SB-Macro-F1) are the primary cell metrics. For patient p , let $C^{(p)}$ be the confusion matrix over a fixed K -class ontology and n_p its number of evaluated cells. Define

$$\bar{C} = \frac{1}{P} \sum_{p=1}^P \frac{C^{(p)}}{n_p}, \quad \text{SB-MacroF1} = \frac{1}{K} \sum_{k=1}^K \frac{2\bar{C}_{kk}}{\sum_j \bar{C}_{kj} + \sum_i \bar{C}_{ik}}. \quad (1)$$

A zero denominator contributes zero. Each patient’s entire matrix, not each class row, is normalized to unit mass. This differs from cell-weighted pooled Macro-F1 and from mean within-subject Macro-F1. We retrospectively re-score the frozen OOF predictions under this primary reporting convention; checkpoints are not reselected. Historical cell-weighted scores remain in Table 17.

Main-table intervals use 2,000 paired patient-bootstrap resamples, with the same patient multiplicities across models. They are pointwise, conditional on the fitted models and fixed fold design, and do not include model-selection or training-seed uncertainty. The main comparison is descriptive; an interval for an individual score is not a test of pairwise superiority. Historical cell-weighted tests against HCE and XGBoost remain separately documented and are not transferred to the new metric. Exact-path accuracy, subtype-tree distance, root-excluded hierarchical F1, and consistency are appendix diagnostics.

Patient characterization uses one prediction per patient, with patient-level accuracy, macro-F1, and macro-AUROC. Its effective disease sample is 40, not 361,792. Nested model selection prevents direct test exposure but does not remove confounding or uncertainty from exploratory analysis choices.

The paired-RNA probe also uses subject-balanced pooled Macro-F1, but has a different training budget and preprocessing protocol; its scores must not be interpreted as additional architecture rows in the main table. Appendix C.8 gives the definitions.

3.4 SUPPORTING RELIABILITY ANALYSES

The paired-RNA analysis is an internal reconstruction diagnostic, since RNA also supplies the reference annotations. In the current matched-input probe, HVG selection is fitted separately within inner and outer training subsets; this prevents feature-selection exposure to their evaluation patients, but does not make RNA an independent validation of RNA-derived labels. Centering, patient-identity prediction, and recorded QC controls probe alternative explanations for disease association. Feature-panel and inference-cell subsampling are secondary redundancy diagnostics, not evidence of reduced training requirements or clinically efficient assays.

Subject-cell allocation experiments are preserved in a standalone supporting report (`sampling_report.tex`), not included as a contribution of the present paper. Their external RNA cohorts do not validate APT cell typing or APT disease prediction.

4 BENCHMARK REFERENCE MODELS

APT-only comparisons. LR and XGBoost provide linear and nonlinear tabular references. Flat and HCE Transformers and a Soft-cascade Transformer provide backbone-matched neural recipes. All use APT at inference and the same outer patient partitions. Training and validation follow each recipe’s documented protocol; identical outer folds do not imply identical training-set sizes or fully matched tuning. Linear SVM, Random Forest, and reference correlation remain in the full appendix inventory. A full-budget plain MLP/ResNet evaluation is not available; its absence is a coverage limitation, not evidence of incompatibility.

Hierarchy-aware reference. Soft-cascade Transformer embeds aptamer identities and values, and uses the final [CLS] representation z for cell prediction. A coarse head provides probabilities $p(m \mid x)$ for the five lineages. For subtype k with parent $m(k)$, its logit combines a global prediction with a softly weighted lineage expert:

$$s_k(x) = g_k(z) + \alpha p(m(k) \mid x) e_{m(k),k}(z) + \gamma \log\{\epsilon + (1 - \epsilon)p(m(k) \mid x)\}. \quad (2)$$

The independent coarse and fine heads are evaluated without constrained decoding. Flat removes the cascade; HCE instead uses a two-level hierarchical loss adaptation (Cultrera di Montesano et al., 2026). Only Soft-cascade uses cross-disease supervised contrastive pairing, so their objectives and auxiliary disease-label usage differ. These are reference-recipe comparisons, not component ablations. No routing or contrastive contribution is claimed in isolation. Appendix B gives the full architecture, losses, hyperparameters, and component matrix.

Input compatibility. Gene-based pretrained models such as scGPT and Geneformer (Cui et al., 2024; Theodoris et al., 2023) do not directly accept anonymous aptamer feature IDs without a verified mapping or a separately trained adapter. Supplying paired RNA would change the input contract. This distinguishes an undefined direct APT input from an adaptable but unevaluated model; it does not justify claiming superiority over transcriptomic annotation methods.

Secondary analyses. A separate matched-input probe varies APT and RNA while holding its training budget and patient folds fixed. The patient-level case study uses disease-label-free XGBoost cell predictions and nested cross-fitting: disease-training compositions come from inner held-out patients, and disease-test compositions come from the untouched outer fold. Downstream LR selection remains within development patients. These analyses contextualize APT typing; they are not additional primary benchmark tasks or validated clinical endpoints.

Task	Model	Patient-disjoint	Cell split	Inflation
Coarse lineage	XGBoost	0.322	0.372	+0.050
	Soft-cascade Transformer	0.326	0.417	+0.090
Fine subtype	XGBoost	0.143	0.212	+0.069
	Soft-cascade Transformer	0.152	0.260	+0.108

Table 1: Historical cell-weighted pooled Macro-F1 under patient-disjoint evaluation versus the available cell-level split controls. All four model-resolution comparisons are higher under cell splitting. These are protocol comparisons, not matched-seed causal estimates of leakage alone; no uncertainty for the protocol difference is asserted. Differences were computed before rounding. The separate APT-only leaderboard (Table 2) reweights patients and therefore uses a different metric.

Model	Input	Coarse SB-Macro-F1 $\uparrow$	Fine SB-Macro-F1 $\uparrow$
Logistic Regression	APT	0.298 [0.281, 0.315]	0.126 [0.107, 0.139]
XGBoost	APT	0.325 [0.302, 0.345]	0.137 [0.120, 0.149]
Flat Transformer	APT	0.326 [0.300, 0.348]	0.139 [0.114, 0.155]
HCE Transformer	APT	0.335 [0.310, 0.356]	0.143 [0.121, 0.160]
Soft-cascade Transformer	APT	0.332 [0.306, 0.354]	0.145 [0.122, 0.160]

Table 2: Primary APT-only cell-typing comparison: five patient-disjoint folds, 40 patients, 361,792 cells. Entries are subject-balanced pooled Macro-F1 with pointwise 95% patient-bootstrap intervals (2,000 resamples). Frozen predictions are retrospectively re-scored with equal total patient weight; checkpoints are unchanged. Comparisons are descriptive, conditional on fitted models. Transformer auxiliary supervision differs (Table 3); full-budget MLP coverage remains unavailable.

5 RESULTS

5.1 CELL-LEVEL SPLITTING GIVES AN OPTIMISTIC EVALUATION

The benchmark first asks whether evaluation on additional cells represents generalization to additional patients. Table 1 contrasts the historical cell-level split controls with the frozen patient-disjoint evaluation. Cell-level splitting gives higher scores for both models at both resolutions: +0.050/ +0.069 coarse/fine Macro-F1 for XGBoost and +0.090/ +0.108 for Soft-cascade. The apparent fine advantage of Soft-cascade over XGBoost is also larger under cell splitting (0.048 versus approximately 0.009 under patient-disjoint evaluation).

These comparisons use the historical cell-weighted metric in both columns; they must not be compared numerically with the subject-balanced main leaderboard below. They contrast evaluation protocols, including their training/test allocation, rather than estimating a causal leakage effect under a fully matched randomized design. They support a specific conclusion: held-out-cell performance should not be presented as evidence of new-patient generalization. They do not show that all model rankings reverse or quantify optimism for every possible cohort.

5.2 REFERENCE PERFORMANCE UNDER THE PATIENT-DISJOINT CONTRACT

Table 2 compares APT-only reference predictions using subject-balanced pooled Macro-F1 on the same 40 held-out patients. Coarse/fine scores are 0.298/0.126 for LR and 0.325/0.137 for XGBoost. Flat, HCE, and Soft-cascade Transformers reach 0.326/0.139, 0.335/0.143, and 0.332/0.145. These point estimates establish reference performance, not a statistically demonstrated cascade advantage.

The complete inventory and historical cell-weighted scores are in Tables 16 and 17. Fine prediction remains difficult across references. However, five-lineage and 27-subtype tasks differ in class count, prevalence, and annotation boundaries; their score difference is not an isolated assay-resolution effect. Full class-level and hierarchy diagnostics are in Appendix C.7. Missing full-budget MLP coverage and objective differences between Transformer recipes limit conclusions about architecture choice.

5.3 SECONDARY CONTEXT: RNA AND PATIENT SUMMARIES

The paired-input diagnostic is not an extension of the APT-only leaderboard. Under its capped matched-probe protocol, an MLP reaches coarse/fine SB-Macro-F1 0.310/0.094 from APT and 0.871/0.584 from RNA; APT+RNA reaches 0.873/0.606. Training-subset-only HVG selection avoids feature-selection exposure, but RNA also supplies the reference annotations. The comparison therefore concerns internal label reconstruction, not independent biological validation or APT algorithm superiority (Appendix C.2).

As a secondary patient-level case study, nested subtype summaries reduce reference-composition error from 0.0231 for a training-mean baseline to 0.0148 (paired reduction 0.0084; 95% patient-bootstrap CI [0.0065, 0.0101]). Their disease Macro-F1 is 0.593 versus 0.477 for lineage summaries, but the difference is unresolved (+0.116; [-0.015, 0.262]). Strong direct APT disease association is sensitive to patient-wise centering and cannot be separated from acquisition effects with the available metadata. Forty patients do not establish clinical validity. The complete case study and controls remain in Appendices C.6 and C.5, rather than forming a second headline contribution.

6 DISCUSSION

The central lesson of APT-Bench is that the unit of generalization must be part of the cell-typing benchmark. The observed cell-split optimism is consistent across the two evaluated model families and both resolutions, but its magnitude is specific to the available protocol comparisons. Patient-disjoint testing and subject-balanced scoring address different issues: the former separates people across training and test sets, while the latter limits unequal cell recovery from dominating the score.

Benchmark contribution versus model contribution. The resource specifies an APT input, operational ontology, fixed patient folds, metrics, and reference results. Soft-cascade is a hierarchy-aware reference, not a claimed algorithmic advance. The value of the benchmark must therefore be judged by the relevance of its task, evaluation validity, and usability, not by whether that reference wins. Complete data access, permissions, annotation provenance, and a portable evaluator remain necessary release requirements; incomplete delivery cannot be repaired by stronger novelty language. Full-budget MLP coverage and more comprehensive APT-compatible annotation baselines also remain limitations.

Boundaries of the evidence. The APT cohort contains 40 participants and RNA-derived reference labels. Independent annotation checks and additional APT cohorts are needed to assess broader generalization. Patient-disjoint testing does not remove disease-correlated acquisition effects, so the secondary disease study is not clinical validation. Paired RNA contextualizes label reconstruction rather than establishing an aptamer information limit. Subject-cell sampling analyses are preserved in a separate report; neither a scaling law nor a resource-allocation algorithm is a contribution of this paper.

7 CONCLUSION

APT-Bench specifies patient-disjoint cell typing from single-cell aptamer profiles and establishes classical and neural reference results under an explicit patient-aware metric. The evaluated cell-level split controls give consistently higher performance than the new-patient protocol, demonstrating why held-out cells must not substitute for held-out patients in reporting generalization. The benchmark's next requirements are complete reproducible release and independent APT validation, not an unsupported claim of algorithmic or clinical superiority.

AI USE STATEMENT

Generative-AI assistance through ChatGPT and Codex was used in the revision workflow for research-design suggestions, literature search, manuscript editing, analysis-code development, debugging, experiment orchestration, and scientific figure preparation. AI-generated suggestions were not themselves experimental evidence; reported computational results are tied to saved run artifacts. **TODO: Authors: confirm the complete tool/model inventory and versions, describe the human**

checks actually performed on code, citations, figures, and statistical claims, and affirm responsibility for the final submission. Automated checks do not substitute for this author verification.

ETHICS STATEMENT

APT-Bench is intended for research evaluation, not clinical decision-making. The small single-cohort sample and unresolved acquisition effects preclude claims of clinical validity or demographic fairness. Strong patient-specific signals also create a privacy concern: removing direct identifiers alone does not establish that cell profiles cannot be linked to their source individuals. Any data release must respect participant consent and applicable access terms. **TODO: Data owners: supply the actual ethics approval or exemption and consent terms; confirm de-identification procedures, permitted sharing/access controls, and conflicts of interest. These facts have not been verified from the available analysis artifacts; preserve reviewer anonymity when providing documentation.**

REPRODUCIBILITY STATEMENT

The repository records the fixed five-fold partition in `benchmark/splits/patient_folds.json`, selected artifact checksums and structural audits in `benchmark/release_audit/`, and artifact-level reproduction commands in `REPRODUCIBILITY.md`. The latter distinguishes re-building figures/PDFs from rerunning models. Per-experiment protocol files record sampling settings and seeds; the controlled simulation includes per-replicate results and source hashes. A snapshot of the local analysis environment is provided, but is not a lockfile for all historical training runs. **TODO: Finalize anonymous reviewer access, data permissions and licenses, archival versioning, original training environments, and a portable raw-data-to-results evaluator. The present repository is not yet a complete independently reproducible benchmark release.**

AUTHOR CONTRIBUTIONS

TODO: Add author contributions using CRediT roles (for example: conceptualization, methodology, software, validation, formal analysis, data curation, visualization, writing, supervision, and funding acquisition). Preserve anonymity in the review version if required.

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TODO: Add funding sources and grant numbers, computational resources, data or clinical contributors, and non-author assistance; verify whether any acknowledgment must be omitted or anonymized during double-blind review.

REFERENCES

- Dvir Aran, Agnieszka P Looney, Leqian Liu, Esther Wu, Valerie Fong, Austin Hsu, Suzanna Chak, Ram P Naikawadi, Paul J Wolters, Adam R Abate, et al. Reference-based analysis of lung single-cell sequencing reveals a transitional profibrotic macrophage. *Nature immunology*, 20(2):163–172, 2019.
- Joshua D Cohen, Lu Li, Yuxuan Wang, Christopher Thoburn, Bahman Afsari, Ludmila Danilova, Christopher Douville, Ammar A Javed, Fay Wong, Austin Mattox, et al. Detection and localization of surgically resectable cancers with a multi-analyte blood test. *Science*, 359(6378):926–930, 2018.
- Haotian Cui, Chloe Wang, Hassaan Maan, Kuan Pang, Fengning Luo, Nan Duan, and Bo Wang. scgpt: toward building a foundation model for single-cell multi-omics using generative ai. *Nature methods*, 21(8):1470–1480, 2024.
- Sebastiano Cultrera di Montesano, Davide D’Ascenzo, Srivatsan Raghavan, Ava P Amini, Peter S Winter, and Lorin Crawford. Improving atlas-scale single-cell annotation models with hierarchical cross-entropy loss. *Nature Computational Science*, 6(3):243–249, 2026.
- CZI Cell Science Program, Shibla Abdulla, Brian Aevermann, et al. CZ CELLxGENE Discover: a single-cell data platform for scalable exploration, analysis and modeling of aggregated data. *Nucleic Acids Research*, 53(D1):D886–D900, 2025. doi: 10.1093/nar/gkae1142.

- Nick Erickson, Lennart Purucker, Andrej Tschalzev, David Holzmüller, Prateek Desai, David Salinas, and Frank Hutter. Tabarena: A living benchmark for machine learning on tabular data. *Advances in Neural Information Processing Systems*, 38, 2026.
- Yury Gorishniy, Ivan Rubachev, Valentin Khrulkov, and Artem Babenko. Revisiting deep learning models for tabular data. In *Advances in Neural Information Processing Systems*, volume 34, pp. 18932–18943, 2021.
- Léo Grinsztajn, Edouard Oyallon, and Gaël Varoquaux. Why do tree-based models still outperform deep learning on tabular data? arxiv. *arXiv preprint arXiv:2207.08815*, 2022.
- Minsheng Hao, Jing Gong, Xin Zeng, Chiming Liu, Yucheng Guo, Xingyi Cheng, Taifeng Wang, Jianzhu Ma, Xuegong Zhang, and Le Song. Large-scale foundation model on single-cell transcriptomics. *Nature methods*, 21(8):1481–1491, 2024.
- Yuhan Hao, Stephanie Hao, Erica Andersen-Nissen, William M Mauck, Shiwei Zheng, Andrew Butler, Maddie J Lee, Aaron J Wilk, Charlotte Darby, Michael Zager, et al. Integrated analysis of multimodal single-cell data. *Cell*, 184(13):3573–3587.e29, 2021. doi: 10.1016/j.cell.2021.04.048.
- Shangru Jia, Artem Lysenko, Keith A Boroevich, Alok Sharma, and Tatsuhiko Tsunoda. schdeepinsight: a hierarchical deep learning framework for precise immune cell annotation in single-cell rna-seq data. *Briefings in Bioinformatics*, 26(5):bbaf523, 2025.
- Leila C Kahwati, Matthew Avenarius, Leslie Brouwer, Norah L Crossnohere, Chyke A Doubeni, Cecelia Miller, Mariam Siddiqui, Christiane Voisin, Roberta C Wines, and Daniel E Jonas. Multicancer detection tests for screening: a systematic review. *Annals of Internal Medicine*, 178(11):1591–1604, 2025.
- Kasia Z Kedzierska, Lorin Crawford, Ava P Amini, and Alex X Lu. Zero-shot evaluation reveals limitations of single-cell foundation models. *Genome Biology*, 26(1):101, 2025.
- Chunlei Liu, Sichang Ding, Hani Jieun Kim, Siqun Long, Di Xiao, Shila Ghazanfar, and Pengyi Yang. Multitask benchmarking of single-cell multimodal omics integration methods. *Nature Methods*, 22:2449–2460, 2025. doi: 10.1038/s41592-025-02856-3.
- Tianyu Liu, Edward De Brouwer, Archit Verma, Alsu Missarova, Tony Kuo, Nathaniel Diamant, Hanchen Wang, Minsheng Hao, Tommaso Biancalani, Hector Corrada Bravo, et al. Learning multi-cellular representations of single-cell transcriptomics data enables characterization of patient-level disease states. *Cell Systems*, 17(5):101570, 2026. doi: 10.1016/j.cels.2026.101570.
- Malte D Luecken, Scott Gigante, Daniel B Burkhardt, Robrecht Cannoodt, Daniel C Strobl, Nikolay S Markov, Luke Zappia, Giovanni Palla, Wesley Lewis, Daniel Dimitrov, et al. Defining and benchmarking open problems in single-cell analysis. *Nature Biotechnology*, 43:1035–1040, 2025. doi: 10.1038/s41587-025-02694-w.
- Christopher S McGinnis, Lyndsay M Murrow, and Zev J Gartner. Doubletfinder: Doublet detection in single-cell rna sequencing data using artificial nearest neighbors. *Cell Systems*, 8(4):329–337.e4, 2019. doi: 10.1016/j.cels.2019.03.003.
- Vladimir Shitov, Mohammad Moghareh Dehkordi, and Malte D Luecken. Benchmarking sample representations from single-cell data: Metrics for biologically meaningful embeddings. In *ICLR Workshop on Learning Meaningful Representations of Life*, 2025. URL <https://openreview.net/forum?id=IoRv5afWtb>.
- Christina V Theodoris, Ling Xiao, Anant Chopra, Mark D Chaffin, Zeina R Al Sayed, Matthew C Hill, Helene Mantineo, Elizabeth M Brydon, Zexian Zeng, X Shirley Liu, et al. Transfer learning enables predictions in network biology. *Nature*, 618(7965):616–624, 2023.
- Jonathan CM Wan, Peter Sasieni, and Nitzan Rosenfeld. Promises and pitfalls of multi-cancer early detection using liquid biopsy tests. *Nature Reviews Clinical Oncology*, 22(8):566–580, 2025.

Yu Yang, Xueyan Chen, Jieying Pan, Huiheng Ning, Yaojun Zhang, Yufei Bo, Xianwen Ren, Jiesheng Li, Shishang Qin, Dongfang Wang, et al. Pan-cancer single-cell dissection reveals phenotypically distinct b cell subtypes. *Cell*, 187(17):4790–4811, 2024.

Yong Zeng, Dor D Abelman, Althaf Singhawansa, Nicholas Cheng, Yuanchang Fang, Sasha C Main, Emma Bell, Wenbin Ye, Ping Luo, Samantha L Wilson, et al. A pan-cancer compendium of 1,294 plasma cell-free dna methylomes and fragmentomes enabling multicancer detection. *Nature Cancer*, 7(2):384–398, 2026.

Yuanhang Zhang, Xiaorui Zhou, Yu Zhong, Xi Chen, Zeyu Li, Rui Li, Pengfei Qin, Shanshan Wang, Jianhua Yin, Shang Liu, et al. Pan-cancer scrna-seq analysis reveals immunological and diagnostic significance of the peripheral blood mononuclear cells. *Human Molecular Genetics*, 33(4):342–354, 2024.

A APPENDIX

B TRANSFORMER REFERENCE IMPLEMENTATION DETAILS

This appendix gives the implementation details for the hierarchy-aware reference ladder introduced in Section 4: a flat Transformer, its HCE counterpart, and Soft-cascade Transformer. The cascade focuses on the cell-level hierarchy, lineage $\rightarrow$ subtype, while disease-association audits use separate tabular baselines and patient-level aggregation. The released code retains the historical identifier `DropCascade` for artifact compatibility. That identifier denotes a reproducible benchmark baseline, not a claim that its global path, auxiliary coarse objective, soft routing, or contrastive construction independently improves performance.

B.1 CELL ENCODER

Each cell is represented by a standardized APT vector $x \in \mathbb{R}^{293}$. We embed feature identity and feature value separately. For aptamer index i , we form a token

$$t_i = \text{Emb}(i) + \text{MLP}_{\text{val}}(x_i), \quad i = 1, \dots, 293. \quad (3)$$

A learnable `[CLS]` token is prepended to the resulting sequence and processed by a pre-norm Transformer encoder with 4 layers, 8 attention heads, hidden dimension 256, feed-forward dimension 512, GELU activations, and dropout 0.1. The final `[CLS]` representation

$$z = \text{Encode}(x) \in \mathbb{R}^{256} \quad (4)$$

serves as the shared cell representation for downstream tasks.

B.2 SOFT-CASCADE REFERENCE

Let $M = 5$ be the number of coarse lineages and $K = 27$ the number of fine subtypes. Each subtype k belongs to exactly one parent lineage $m(k)$. A coarse head produces lineage logits

$$c = W_c z \in \mathbb{R}^M, \quad (5)$$

with routing distribution

$$p(m \mid x) = \text{softmax}(c/\tau)_m, \quad (6)$$

where $\tau = 1$.

Instead of a hard cascade, which would force subtype prediction to follow only the top predicted lineage, Soft-cascade Transformer uses a soft mixture of a global subtype path and lineage-specific experts:

$$s_k(x) = \underbrace{g_k(z)}_{\text{global path}} + \alpha p(m(k) \mid x) e_{m(k),k}(z) + \gamma \log(\epsilon + (1 - \epsilon)p(m(k) \mid x)), \quad (7)$$

Here $g(z)$ is a global subtype head, $e_{m,k}(z)$ is the expert associated with lineage m , $\alpha = 1.0$, $\epsilon = 0.1$, and γ is ramped from 0 to 0.5 during early training. The global path retains a direct subtype-logit contribution alongside the routed expert contribution; we do not isolate its effect from the other components. Final predictions are

$$\hat{k} = \arg \max_k s_k(x), \quad \hat{m} = \arg \max_m c_m. \quad (8)$$

B.3 CROSS-DISEASE CONTRASTIVE REGULARIZATION

The reference model includes a cross-disease supervised contrastive term. Let

$$u = \frac{\text{Proj}(z)}{\|\text{Proj}(z)\|_2} \quad (9)$$

be a normalized projection of the encoder representation. For anchor cell i , positives are cells with the same subtype but a different disease label. Same-subtype cells from the same disease are excluded by construction. Cells from different subtypes are treated as negatives, with additional emphasis on negatives that share the same coarse lineage. The contrastive loss weight is ramped from 0 to 0.2 after the early warmup period. Because we do not compare this construction with standard supervised contrastive learning across all five folds, we make no claim that the cross-disease positive rule independently improves generalization.

B.4 OPTIMIZATION AND MODEL SCOPE

The classification cascade is trained jointly using

$$\mathcal{L} = \lambda_{\text{fine}} \mathcal{L}_{\text{fine}} + \lambda_{\text{coarse}} \mathcal{L}_{\text{coarse}} + w_{\text{contrast}}(t) \mathcal{L}_{\text{contrast}}, \quad (10)$$

where $\mathcal{L}_{\text{fine}}$ and $\mathcal{L}_{\text{coarse}}$ are cross-entropy losses with $\lambda_{\text{fine}} = 1.0$ and $\lambda_{\text{coarse}} = 0.5$. The contrastive weight increases to 0.2 according to the schedule in §B.3.

We optimize the cascade using AdamW with learning rate 10^{-4} and weight decay 10^{-5} , gradient clipping at norm 1.0, batch size 512, and mixed-precision training. Models are trained for at most 50 epochs with early stopping based on validation fine-level macro-F1 with patience 10. The final configuration does not use class-balanced sampling because it underperformed natural sampling after matching the number of optimization steps; the single-split development diagnostic is provided in Appendix C.11.

Backbone-matched Transformer references. The flat FT-style Transformer reference (Gorishniy et al., 2021) removes the cascade while retaining the identical APT tokenizer, Transformer dimensions, optimizer, early stopping, and training budget above. We call it FT-style because its tokenizer is exactly the APT feature tokenizer used by Soft-cascade Transformer, rather than the original numerical/categorical FT-Transformer tokenizer. This model and its HCE counterpart predict the same $M + K$ lineage-and-subtype nodes. The flat model applies independent cross-entropy to the lineage and subtype logit slices. For HCE (Cultrera di Montesano et al., 2026), let $R_{ij} = 1$ when node j is node i or its descendant, $p = \text{softmax}(q)$ over all nodes, and $s = Rp$. We minimize $-\lambda_{\text{fine}} \log s_{M+k} - \lambda_{\text{coarse}} \log s_m$ for observed subtype k and lineage m . Because every fine label here is a leaf, the fine HCE term alone would reduce to ordinary CE; including the simultaneously observed parent label makes this a stated two-level adaptation rather than an unqualified reproduction of the mixed-granularity setting in the original work. All three Transformer models are evaluated from the checkpoint selected by patient-validation fine macro-F1, without an additional post-selection calibration stage. Table 3 makes the remaining objective differences explicit. In particular, only the soft-cascade reference uses disease labels to construct cross-disease contrastive pairs; Flat and HCE use no disease labels and no contrastive term. The ladder therefore holds backbone capacity, optimization budget, folds, and checkpoint selection, but it is not an objective-matched component ablation. We do not attribute any score difference to soft routing, hierarchical loss, or contrastive regularization in isolation.

Component	Flat FT-style Transformer	FT-style Transformer + HCE	Soft-cascade Transformer
APT tokenizer and Transformer dimensions	Same	Same	Same
Patient folds, optimizer, budget, checkpoint rule	Same	Same	Same
Fine-label supervision	Independent CE	Leaf HCE NLL	CE
Coarse-label supervision	Independent CE	Ancestor HCE NLL	CE
Cross-disease supervised contrastive term	No	No	Yes ($w \leq 0.2$)
Disease labels used by cell model	No	No	Contrastive pairing only
Global subtype path and lineage experts	No	No	Yes
Soft lineage routing	No	No	Yes

Table 3: Components of the three neural references. “Same” denotes an identical specification, not shared fitted weights. The models match the tokenizer, encoder dimensions, data folds, optimizer, maximum epochs, early stopping rule, and checkpoint criterion, but they do not match the complete training objective. In particular, disease labels enter only the soft-cascade contrastive pairing rule. Consequently, comparisons among these rows evaluate complete recipes and are not component ablations.

Disease recognition is not handled by a dedicated head inside Soft-cascade Transformer. The reference recipe addresses cell-level hierarchy prediction; disease-source prediction is evaluated separately as a within-cohort audit. High disease-source scores do not establish biological attribution or external clinical validity.

Strictly nested cell-output transfer audit. For each disease outer fold f , we train a disease-label-free XGBoost subtype classifier on all 32 outer-development patients and predict the cells of the eight untouched patients in f . Disease-training features require a second cross-fit: for each of the other four fixed folds g , a separate subtype model is trained on the 24 patients outside $f \cup g$ and predicts fold g . We average the resulting 27-class probability vectors within patient to obtain soft subtype composition; soft lineage composition sums subtype probabilities under the frozen subtype-to-lineage map. A standardized, class-weighted multinomial LR is fit to the 32 inner-OOF patient vectors, with $C \in \{0.01, 0.1, 1, 10, 100\}$ selected by patient-level inner CV, and evaluated once on the eight outer-test vectors. Raw proportions are primary; CLR with a prespecified 10^{-6} pseudocount is a sensitivity analysis. All uncertainty and permutation analyses resample or shuffle patients, never cells. To test conditional rather than marginal subtype value, we additionally compare three nested feature sets under the same folds and inner selection: recorded-QC summaries alone, QC plus the five predicted lineage proportions, and QC plus the 27 predicted subtype proportions. The last representation contains lineage implicitly because child probabilities sum to their parents.

The main cell-identity results are reported in Table 2; the following sections supply their implementation and diagnostic details.

C DATASET AND EVALUATION DETAILS

C.1 DATASET CONSTRUCTION AND DOCUMENTATION CHECKLIST

The available study workflow establishes the high-level sequence used to construct APT-Bench: clinical peripheral-blood specimens from five cancer groups and healthy controls are converted to single-cell suspensions; cell-profile and viability quality control precedes a paired single-cell transcriptome and 293-aptamer assay; separate transcriptome and aptamer-barcode libraries are linked at the cell level; and transcriptome-derived clusters and annotations provide the cell labels. For a benchmark release, this overview must be accompanied by the following provenance and protocol fields.

Outstanding disease-confounding validation. **TODO:** Obtain a sample-linked acquisition manifest from the data owner: patient and sample IDs, collection and processing dates, APT assay batch, plate/pool, reagent lot, library-preparation batch, sequencing run, site, and available fresh/frozen and handling records. Audit disease-by-batch overlap before choosing a validation design. Where disease groups have adequate cross-batch support, evaluate patient-disjoint, held-out-batch generalization with all preprocessing and selection restricted to training data. If disease and batch are inseparable, obtain crossed-batch samples or an independent APT cohort; statistical adjustment alone cannot identify their separate contributions. The NC matrices must not be treated as external vali-

dation if they are the healthy subset of this cohort. These validations are pending, not completed. Expression and QC matrices alone do not establish acquisition provenance; current disease results remain within-cohort associations.

Transcriptomic preprocessing and cell quality control. The transcriptomic count matrix was processed with Seurat following its standard workflow of normalization, feature selection, dimensionality reduction, neighborhood-graph construction, and clustering (Hao et al., 2021). DoubletFinder was used to identify and remove predicted doublets (McGinnis et al., 2019). Low-quality cells were additionally filtered using per-cell UMI count, detected-gene count, and mitochondrial-transcript fraction. These steps preceded cluster annotation and construction of the linked APT benchmark cohort. **TODO: Report the Seurat and DoubletFinder versions, every parameter and random seed, the exact UMI/gene/mitochondrial thresholds, and cell counts before and after each filter.**

Manual annotation and reference-label reliability. After Seurat clustering, investigators manually assigned lineage and subtype names by inspecting canonical marker-gene expression reported in prior PBMC and pan-cancer studies (Zhang et al., 2024; Yang et al., 2024). No quantitative annotation confidence, inter-annotator agreement, or external reference-transfer score is available. Accordingly, these labels are operational reference annotations, not error-free biological ground truth, and benchmark scores are conditional on this manual taxonomy. The *Unknown* label was assigned when a cluster did not show a coherent canonical marker pattern for the expected PBMC subtypes; this was a manual reject decision rather than a calibrated confidence threshold. Unknown cells are excluded from the primary 27-subtype benchmark. **TODO: Provide the full subtype-by-marker citation table, the number and expertise of annotators, whether they were blinded to disease, the adjudication procedure, and the explicit rule used to assign Unknown.**

Cross-disease semantics and hierarchy mapping. The same 27 subtype labels and definitions were applied across all six disease groups; no subtype was renamed or redefined for a particular disease. This gives the labels a consistent operational meaning, but it does not prove that marker expression or biological state is invariant across diseases. Each subtype is assigned to one of five parent lineages by a manually curated lookup. The exported lookup and structural audit are available under `benchmark/release_audit/`: the existing subtype summary has 27 unique subtypes, five parents, and 361,792 cells, and the fold manifest has five disjoint groups of eight patients. These checks establish internal bookkeeping consistency, not independent biological validity. **TODO: Document independent validation of the subtype-to-lineage mapping, including reviewers, disagreements, adjudication, and any marker-based checks.**

C.2 PAIRED-INPUT INTERNAL RECONSTRUCTION REFERENCE

We use paired RNA as an internal label-reconstruction reference, separate from the primary APT-only model comparison. The current matched-input experiment replaces the earlier data-development HVG representation: 2,000 features are selected by mean-binned log-count dispersion within each inner or outer training subset, using the original RNA count matrix. All 361,792 benchmark cells are aligned by barcode. This controls feature-selection exposure to evaluation patients, but the transcriptomic modality also underlies the manual annotations. Stronger RNA reconstruction therefore does not independently validate those annotations or establish an intrinsic APT information limit.

Within each of the five fixed patient-disjoint folds, we compare five inputs: APT only, paired RNA only, APT+RNA, RNA plus random 293-dimensional noise, and RNA plus APT vectors shuffled within patient. The noise control checks feature dimension, while the within-patient shuffle preserves patient-level APT structure but breaks cell-level RNA-APT pairing. All inputs use the identical deterministic sample of up to 1,000 training cells per development patient and all cells from the eight untouched test patients. RNA features are selected as 2,000 highly variable genes using only the current training fold; all features are standardized using training-fold statistics. We fit patient- and class-weighted linear SGD and one-hidden-layer MLP probes for 40 epochs at both resolutions, using one inner patient-disjoint validation fold to select the regularization strength. Scores use subject-normalized OOF confusion matrices, and 2,000 paired patient-bootstrap replicates quantify

modality contrasts. This matched-budget probe uses the main subject-balanced metric, but is not an architecture comparison with the full-training model ladder in Table 2.

Probe	Input	Coarse Macro-F1	Fine Macro-F1
SGD	APT only	0.277 [0.254, 0.296]	0.090 [0.076, 0.098]
SGD	RNA only	0.742 [0.731, 0.750]	0.431 [0.387, 0.453]
SGD	APT + RNA	0.751 [0.740, 0.762]	0.458 [0.413, 0.483]
SGD	RNA + noise	0.742 [0.732, 0.751]	0.409 [0.371, 0.430]
SGD	RNA + shuffled APT	0.747 [0.732, 0.760]	0.448 [0.405, 0.471]
MLP	APT only	0.310 [0.292, 0.325]	0.094 [0.081, 0.103]
MLP	RNA only	0.871 [0.854, 0.883]	0.584 [0.529, 0.610]
MLP	APT + RNA	0.873 [0.858, 0.884]	0.606 [0.551, 0.635]
MLP	RNA + noise	0.870 [0.855, 0.880]	0.571 [0.525, 0.593]
MLP	RNA + shuffled APT	0.876 [0.862, 0.887]	0.596 [0.544, 0.622]

Table 4: Paired-input internal reconstruction diagnostic under the five fixed patient-disjoint folds. Each input uses the same deterministic sample of up to 1,000 training cells per development patient, training-fold standardization, all held-out cells, patient- and class-weighted training, and one inner patient-disjoint validation fold for regularization. Brackets are 2,000 paired patient-bootstrap intervals for subject-balanced pooled Macro-F1. RNA features are selected separately within inner and outer training subsets. This prevents feature-selection exposure to their evaluation patients, but the RNA-derived operational annotations are not independently validated by RNA reconstruction.

For the MLP probe, paired RNA improves coarse/fine macro-F1 over APT by +0.562 and +0.490. The concatenated representation improves fine macro-F1 over RNA alone by +0.022 ([0.013, 0.037]), and exceeds RNA plus shuffled APT by +0.010 ([0.002, 0.017]). The linear SGD probe shows the same direction for RNA+APT versus RNA (+0.028; [0.013, 0.041]), but its shuffled-APT contrast is unresolved (+0.010; [-0.005, 0.025]). RNA therefore reconstructs the operational taxonomy far better than APT, while the small combined-input gain should be read as internal complementary association rather than an estimate of annotation accuracy or a clinical claim.

- **TODO: Report recruiting site(s) and dates, prospective versus retrospective collection, exact patient count per group, inclusion/exclusion and diagnostic criteria, and whether each participant contributed exactly one specimen.**
- **TODO: Report available demographics and clinical covariates, including age, sex, disease stage, treatment status, and comorbidities; summarize missingness and state which fields cannot be released.**
- **TODO: Report blood volume, collection tubes, time to processing, cell-isolation and cryopreservation procedures, quality-control thresholds, and sample/cell exclusion counts.**
- **TODO: Report assay/platform and reagent versions, sequencing instruments and depth, 293-probe panel design, and a verified protein-target manifest or explicitly state why only anonymized probe identifiers can be released.**
- **TODO: Report barcode-matching rates, APT normalization and filtering, all remaining software versions and parameters, and whether annotation was performed before investigators examined APT profiles or disease labels.**
- **The documented final cell-flow accounting is 374,854 linked post-QC cells – 13,062 Unknown cells = 361,792 benchmark cells. TODO: Data owners: provide participant and cell counts at all earlier collection/QC exclusions before a complete cohort-flow diagram can be drawn. Supply the approval, consent, access, and licensing evidence listed in the Ethics and Reproducibility Statements.**

C.3 CHANCE-ADJUSTED RESOLUTION DIAGNOSTIC

Raw coarse and fine Macro-F1 compare five-class and 27-class tasks with different label frequencies and decision boundaries. We therefore treat their difference as descriptive, not evidence of a

resolution deficit beyond task cardinality. Using the released 40 patient-specific OOF confusion matrices from the matched paired-input diagnostic, we normalize each matrix to unit mass and average over patients to obtain M . Let $p_k = \sum_j M_{kj}$ and $q_k = \sum_i M_{ik}$. The independence reference $M_{ij}^0 = p_i q_j$ retains both true and predicted marginals while breaking their association. We report

$$S_0 = \frac{1}{K} \sum_k \frac{2p_k q_k}{p_k + q_k}, \quad S_{\text{adj}} = \frac{\text{MacroF1}(M) - S_0}{1 - S_0}, \quad (11)$$

with zero contribution when $p_k + q_k = 0$. This is F1 of an independence matrix, not the exact expectation of finite-sample permutation F1. Each of 2,000 paired patient-bootstrap draws recomputes M , both marginals, and the adjustment; the same patient draw is used for both resolutions and all modalities. These intervals condition on the fitted models and do not include retraining variation.

Probe	Input	Resolution	Raw F1	Chance ref.	Adjusted F1 [95% interval]
SGD	APT	coarse	0.277	0.185	0.113 [0.093, 0.134]
SGD	APT	fine	0.090	0.033	0.059 [0.046, 0.067]
SGD	RNA	coarse	0.742	0.197	0.678 [0.666, 0.689]
SGD	RNA	fine	0.431	0.035	0.410 [0.367, 0.433]
SGD	APT + RNA	coarse	0.751	0.198	0.690 [0.676, 0.702]
SGD	APT + RNA	fine	0.458	0.035	0.438 [0.393, 0.463]
MLP	APT	coarse	0.310	0.186	0.152 [0.134, 0.171]
MLP	APT	fine	0.094	0.029	0.067 [0.055, 0.076]
MLP	RNA	coarse	0.871	0.200	0.839 [0.819, 0.854]
MLP	RNA	fine	0.584	0.037	0.568 [0.512, 0.596]
MLP	APT + RNA	coarse	0.873	0.200	0.842 [0.824, 0.855]
MLP	APT + RNA	fine	0.606	0.037	0.591 [0.535, 0.621]

Table 5: Marginal-chance adjustment for the paired-input reconstructability diagnostic. Scores use subject-balanced pooled confusion matrices from the same 40 OOF patients. Each of 2,000 paired patient-bootstrap draws recomputes the observed score and null marginals. This is an association-breaking reference, not a task-cardinality, annotation-quality, or within-lineage difficulty control.

For APT, the MLP raw coarse/fine scores 0.310/0.094 have chance references 0.186/0.029 and adjusted scores 0.152/0.067. The difference shrinks from 0.216 to 0.085 (95% interval [0.068, 0.104]). The corresponding SGD adjusted gap is 0.054 ([0.034, 0.078]). Adjusted RNA and APT+RNA differences remain larger for both probes, around 0.25–0.27. The adjustment accounts for marginal chance agreement, but does not control label boundaries, class count, annotation reliability, or within-lineage separability; none of these residual differences establishes a cardinality-independent biological granularity effect.

True-lineage-conditioned subtype decoding is not evaluated here. It requires OOF subtype scores restricted to the children of the true lineage (or separately trained conditional classifiers); hard predictions and confusion matrices cannot recover that decision. In particular, filtering to predictions that already match the true lineage would select easier cases rather than provide this control. Thus the present diagnostic cannot attribute fine errors uniquely to lineage identification versus within-lineage discrimination.

C.4 BENCHMARK INTERFACE AND RELEASE CONTRACT

The primary hierarchy submission interface contains pseudonymous cell and patient IDs, true and predicted lineage, and true and predicted subtype. The disease interface contains one row per patient with the true disease and six class scores. The secondary budget diagnostic additionally records the aptamer measurement budget, inference-time cell budget, selection seed, and the development-only feature ranking. These common interfaces permit new models to be scored without reproducing their internal training pipelines.

The evaluation layer is portable to other clinical single-cell datasets. It implements group-disjoint out-of-fold prediction; patient-clustered bootstrap uncertainty; subject-balanced and cell-weighted pooled coarse and fine Macro-F1, exact-path accuracy, root-excluded hierarchical F1, subtype-tree

(a) Task and evaluation scope				
Resource	Primary input	Reported scope	Main unit	Hierarchy
Open Problems	Multiple	12 living tasks	Task-specific	Task-specific
scMultiBench	RNA/ADT/ATAC	64 real + 22 simulated	Cell/dataset	No
SPARE	scRNA-seq	5 datasets, 8 methods	Sample	No
PaSCient	scRNA-seq	12.5M cells, 2,700 patients	Patient	No
APT-Bench	293 aptamers	1 cohort, 40 patients	Cell + patient	Yes

(b) Protocol components and availability				
Resource	Patient task	Two-axis scale	Assay budget	Public pipeline
Open Problems	Task-specific	No	No	Yes
scMultiBench	No	No	Feature selection	Yes
SPARE	Yes	No	No	Yes
PaSCient	Yes	No	No	Not audited
APT-Bench	Secondary	Separate report	Secondary	Pending release

Table 6: Scope comparison with recent single-cell evaluation resources (Luecken et al., 2025; Liu et al., 2025; Shitov et al., 2025; Liu et al., 2026). “Task-specific” means that the living platform defines multiple contracts rather than one shared clinical protocol. APT-Bench is narrower in cohorts but centers a fixed aptamer modality and patient-disjoint coarse-to-fine cell typing. Patient-level analysis is secondary; sampling studies are in a separate report and do not expand the main contribution. Public release remains a submission requirement, not a completed claim.

distance, the descriptive coarse-minus-fine difference, and sibling versus cross-lineage error; patient-level aggregation versus cell-level pseudoreplication; patient-centering and identity diagnostics; and nested feature- and cell-budget curves. A future dataset can replace the assay features, hierarchy, or clinical endpoint while retaining these interfaces. This modular task–dataset–metric separation follows recent living single-cell benchmarks (Luecken et al., 2025); standardized metadata and programmatic access similarly enable cross-study reuse in CZ CELLxGENE (CZI Cell Science Program et al., 2025).

A complete release must contain the processed APT matrix; pseudonymous cell and patient identifiers; disease and cell labels; hierarchy and feature metadata; the immutable fold manifest; metric implementations; baseline configurations; and a single command that regenerates the headline tables from model predictions. The fold manifest, exported hierarchy, selected checksums, and artifact-rebuild instructions are now recorded in `benchmark/release_audit/` and `REPRODUCIBILITY.md`; these cover selected results, not the complete dataset or training pipeline. **TODO: Before submission, finalize anonymous data/code access, explicit licenses, original training environment locks, an archival DOI, and the end-to-end command. If participant-level data require controlled access, release the complete schema, all labels permitted by consent, the split manifest, evaluator, and access procedure.**

C.5 CONFOUNDING AUDIT OF WITHIN-COHORT DISEASE ASSOCIATION

Exploratory scope and independent sample size. All disease analyses reuse one cohort of 40 independent patients across six groups (6–8 patients per group); the 361,792 cells are repeated observations, not additional independent disease outcomes. Outer-fold disease models are developed on only 32 patients. Nested cross-fitting separates training from evaluation but does not increase the number of independent subjects, remove unmeasured confounding, or turn repeated analyses into independent replications. Regularization and inner validation mitigate model-selection risk without eliminating the uncertainty associated with this small development sample. The representation comparisons and post-hoc sensitivities below are exploratory, not confirmatory evidence of incremental subtype value or clinical utility. Reported patient-bootstrap intervals do not account for the full adaptive history of analysis choices. Clinical validation would require an independent APT cohort with recorded acquisition covariates and a frozen preprocessing, feature-selection, and prediction pipeline; the external scaling experiments do not supply that disease-validation cohort.

Aggregation / Representation	Patient Acc.	Macro-F1	Macro-AUROC	Perm. p
Log cell count + LR [‡]	0.575	0.531	0.800	0.00010
All recorded QC proxies + LR [¶]	0.550	0.566	0.859	–
Nested soft subtype composition + LR [§]	0.600	0.593	0.854	0.00010
Nested soft lineage composition + LR [§]	0.475	0.477	0.766	0.00020
Reference subtype composition + LR [‡]	0.500	0.516	0.776	0.00010
Reference lineage composition + LR [‡]	0.325	0.301	0.743	0.0184

[‡] Patient-level technical or paired-transcriptomic oracle features. [¶] Recovered cell count plus five patient summaries of recorded RNA UMI and detected-gene QC fields; see Appendix C.5. [§] Soft probabilities from disease-label-free XGBoost under strict nested cross-fitting. Composition rows use raw proportions; LR regularization is tuned only within each outer-development set. Composition permutation tests use 10,000 patient-label shuffles.

Table 7: **Associational patient-characterization audit** ($n = 40$). Predicted composition retains disease-associated variation, but its gains over lineage composition, recovered cell count, and recorded RNA-QC proxies have patient-bootstrap intervals spanning zero. These representations may contain biological, patient, or technical structure. Full direct-APT settings and confounding controls appear in Appendix C.5; no row establishes biological mediation or clinical generalization.

Table 7 in the main text summarizes the patient-representation comparison; this section supplies its controls and interpretation limits.

Cell-level prediction of the disease source of a held-out patient. Table 7 reports the exploratory patient-characterization audit. For completeness, Table 9 reports the pooled *cell-level* source-prediction metric, in which every held-out cell is treated as a target and pooled across all 361,792 cells. This is an easier and statistically different task than patient-level characterization, since cells from the same patient are correlated and patients with more cells contribute more weight. We report it only as a confounding diagnostic.

Non-identifiability of disease and acquisition effects. Every participant contributes one acquisition, so patient identity is nested within both disease group and any unrecorded processing batch. The released metadata contain disease and patient identifiers plus transcriptomic UMI and detected-gene counts, but no plate, run, date, library, operator, or site field. Consequently, neither patient-disjoint prediction nor label permutation can identify whether a patient-associated APT offset is biological, technical, or mixed. No analysis below is interpreted as disease biology, biomarker validation, or clinical performance.

Recorded nuisance-only baselines. We aggregate the only available per-cell QC fields to each patient: recovered cell count; medians of $\log(1 + \text{UMI})$ and $\log(1 + \text{detected genes})$; median detected-gene/UMI ratio; and the interquartile ranges of the two log-count variables. Each feature set uses the same outer folds as the composition bridge, four-fold inner patient CV on the 32 outer-development patients to select LR regularization, refitting on all 32, and one evaluation on the sealed eight patients. Table 8 shows that these recorded QC proxies alone approach the predicted-composition result. They remain proxies rather than measured batch labels, so this analysis detects known nuisance structure but cannot rule out unrecorded acquisition effects.

We also test conditional subtype value rather than comparing each representation in isolation. Under the same nested folds and inner regularization selection, recorded QC plus predicted lineage reaches macro-F1 0.583 and AUROC 0.875; recorded QC plus the full predicted subtype vector reaches 0.599 and 0.825. The subtype-minus-lineage macro-F1 increment is +0.016 with 95% paired patient-bootstrap CI $[-0.120, 0.169]$ (10,000 resamples). Thus subtype detail does not provide a resolved conditional increment in this cohort.

Recorded feature set	d	Acc.	Macro-F1	Macro-AUROC
Recovered cell count	1	0.575	0.531	0.800
RNA QC summaries	3	0.475	0.478	0.787
All recorded QC proxies	6	0.550	0.566	0.859
Predicted soft subtype composition	27	0.600	0.593	0.854
Recorded QC + predicted lineage	11	0.575	0.583	0.875
Recorded QC + predicted subtype	33	0.600	0.599	0.825

Table 8: Matched patient-level nuisance audit. All rows use the same outer folds and inner patient-level regularization selection. The predicted-composition minus all-recorded-QC Macro-F1 difference is +0.027 with a 95% paired patient-bootstrap interval of $[-0.157, 0.215]$ (2,000 resamples). Recorded QC fields are incomplete proxies, not batch labels. In the conditional audit, the subtype-minus-lineage Macro-F1 increment is +0.016 with 95% paired patient-bootstrap interval $[-0.120, 0.169]$ (10,000 resamples).

Model	Accuracy	Macro-F1	Weighted-F1	Balanced Acc.
Logistic Regression	0.991	0.989	0.991	0.989
XGBoost	0.990	0.988	0.990	0.987
Linear SVM	0.988	0.986	0.988	0.985
Random Forest	0.963	0.955	0.963	0.953
Reference Correlation [†]	0.921	0.909	0.921	0.915

[†] SingleR-style reference-correlation classifier (Spearman nearest-centroid mapping), following the template-matching principle used by reference-mapping methods such as SingleR (Aran et al., 2019).

Table 9: **Cell-level** prediction of the disease source of a held-out patient’s cells (6-class), pooled over all 361,792 out-of-fold cells from the patient-level 5-fold protocol. This is a different, easier statistical task than the exploratory patient-level characterization reported in Table 7: cells from the same patient are correlated observations rather than independent draws, and this metric implicitly weights patients with more cells more heavily. We report it here only as a cell-level reference point, not as a patient-level diagnostic result.

Patient-wise centering sensitivity. To test whether disease prediction relies on each patient’s absolute expression level rather than only on within-patient relative structure, we repeated Logistic Regression training and evaluation after subtracting each patient’s own per-apramer median from every cell in both training and test data. This transformation preserves within-patient cell-to-cell variation while removing a patient-specific location shift. Cell-level accuracy drops from 0.991 to 0.401 (macro-F1 $0.989 \rightarrow 0.348$), and patient-level majority-vote accuracy drops from 40/40 to 26/40. This collapse shows that disease recognition depends substantially on patient-level absolute-expression structure, but it does not by itself determine whether that structure is biological, technical, or a mixture of both.

Patient-fingerprint diagnostic. We additionally trained a 40-way classifier to predict patient identity from individual cells, using a within-patient 50/50 cell split for this diagnostic only. The classifier identifies the correct patient from a single held-out cell with 88.4% accuracy (macro-F1 0.862), against a 2.5% chance level. Individual cells therefore carry a strong and easily recoverable patient-specific signature, consistent with the centering sensitivity above.

Patient-level summary-feature control. Finally, to confirm that the disease signal does not rely only on cell-level pseudo-replication, we trained a genuinely patient-level classifier using each patient’s median 293-dimensional APT profile. This summary-feature model reaches patient accuracy 0.950, patient macro-F1 0.948, and macro-AUROC 1.000. Together with the sensitivity analyses above, this indicates that disease information is present at the patient level but strongly entangled with absolute-expression offsets that cannot be cleanly separated without acquisition-batch meta-data.

Strict nested cell-output transfer audit. Ordinary pooled OOF predictions are insufficient for stacking because a base model used to create a disease-training feature may have seen the eventual disease outer-test patients. We therefore use the nested protocol defined in Methods: within each outer fold, four XGBoost models trained on 24 patients generate inner-OOF features for the 32 disease-training patients, and a fifth model trained on all 32 predicts the eight outer-test patients. XGBoost receives subtype labels but no disease labels. Soft subtype composition reaches accuracy/macro-F1/AUROC 0.600/0.593/0.854 and soft lineage composition reaches 0.475/0.477/0.766. The paired macro-F1 difference is +0.116 (95% patient-bootstrap CI $[-0.015, 0.262]$). Gold subtype and lineage compositions reach macro-F1 0.516 and 0.301, respectively. The apparently stronger predicted than gold point estimate is not statistically resolved and warns that predicted probabilities may encode model-confidence or technical structure beyond true cell abundance.

This experiment is therefore a representation-transfer audit, not an error- propagation analysis. It does not establish that the transferred signal originates from subtype abundance, that fine subtype is more useful than coarse lineage, that better typing improves patient characterization, or that the signal is independent of confidence and technical structure. Consistent with this distinction, the soft-minus-hard subtype macro-F1 difference is only +0.069 (95% patient-bootstrap CI $[-0.077, 0.221]$), and the predicted-minus-gold subtype difference is +0.077 (95% CI $[-0.100, 0.256]$).

Across 10,000 complete patient-label permutations, predicted subtype and lineage composition have plus-one $p = 0.00010$ and $p = 0.00020$. However, log cell count alone reaches macro-F1 0.531 ($p = 0.00010$), and the predicted-subtype minus cell-count difference is +0.062 (95% CI $[-0.087, 0.218]$). Equal-cell subsampling at 100, 500, and 2,000 cells per patient preserves the qualitative result over 50 seeds. A 50-repeat global-multinomial composition control averages macro-F1 0.170, and fold-ID prediction from global OOF composition is below chance. These controls support a predictive and associational bridge, not causal mediation or independent biological validation.

C.6 PATIENT COMPOSITION RECOVERY

For each of the 40 outer-test patients, we compare the mean soft cell-type probabilities with reference cell fractions. The constant baseline is the equal-patient mean reference composition of the 32 outer-development patients in that fold; no test-patient annotations enter its construction. For K classes and N test patients, the recovery error is

$$E = \frac{1}{N} \sum_{p=1}^N \frac{1}{K} \sum_{k=1}^K |\hat{\pi}_{pk} - \pi_{pk}|.$$

We report $E_{\text{baseline}} - E_{\text{model}}$, so positive values favor predicted compositions. All five lineages or all 27 subtypes are included, without selecting well-recovered classes. Absolute errors across these two resolutions are not directly comparable because their denominators differ. Using 2,000 disease-stratified paired patient-bootstrap replicates, subtype error decreases from 0.02314 to 0.01478, a reduction of 0.00836 ($[0.00648, 0.01010]$). Lineage error changes from 0.05495 to 0.05827, a reduction of -0.00332 ($[-0.01389, 0.00670]$). Intervals are pointwise and conditional on the completed cross-fitted cell models; they do not include training-subset or model-fitting uncertainty. The accompanying per-class audit reports error, signed bias, and rank correlation for every class. Disease-adjusted rank correlations residualize both ranked fractions on disease-group indicators and are descriptive, not evidence of biological specificity or causal independence from disease. Reference recovery does not validate the annotation itself or imply improved disease characterization.

Representation	Legacy pooled OOF	Strict nested
Gold subtype composition	0.572	0.516
Predicted subtype composition	0.400	0.593
Gold lineage composition	0.377	0.301
Predicted lineage composition	0.282	0.477

Table 10: Protocol audit of patient macro-F1. The legacy predicted rows use hard labels from one global pooled-OOF prediction set, CLR features, and fixed LR regularization; the strict rows use nested soft probabilities, raw proportions, and development-only LR selection. Because both feature construction and downstream fitting changed, the numerical differences are not estimates of a leakage effect. Only strict nested results support current claims.

Patient-label permutation control. We evaluated the same patient-level median-profile classifier under 1,000 label permutations. Disease labels were permuted across the 40 patients as indivisible units, preserving the disease-group counts (6, 8, 7, 7, 6, 6); labels were never shuffled across cells. We retained the original five patient folds rather than restratifying after permutation. No permuted dataset reached the observed accuracy of 0.950 or macro-F1 of 0.948, giving plus-one permutation $p = (0 + 1)/(1000 + 1) = 0.000999$ for both statistics. The null accuracy mean was 0.137, with empirical 95% interval [0.025, 0.275]. This rejects a random patient-label explanation under the fixed protocol, but does not distinguish biological signal from patient-associated technical structure.

C.7 PER-SUBTYPE SUPPORT AND PERFORMANCE AUDIT

Figure 2 visualizes the two marginal support-F1 associations for Soft-cascade Transformer and XGBoost. The vertical stack at 40 patients makes the coverage ceiling explicit. These plots and the partial correlations are exploratory: they neither establish causality nor test whether two dependent correlations differ significantly.

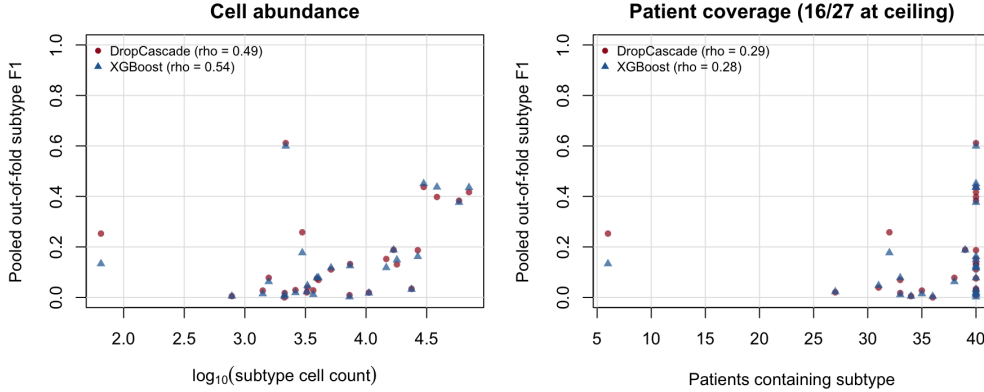


Figure 2: Per-subtype pooled out-of-fold F1 versus cell abundance (left) and patient coverage (right). Each marker is one of 27 subtypes; both models use the same ground-truth support values. Spearman correlations are marginal, and 16 of 27 subtypes occur in all 40 patients.

(a) Support–performance associations

Model	ρ_c	ρ_p	$\rho_{c p}$	$\rho_{p c}$
Soft-cascade Transformer	0.486	0.288	0.411	-0.051
Flat FT-style Transformer	0.505	0.255	0.464	-0.123
FT-style Transformer + HCE	0.513	0.318	0.427	-0.034
XGBoost	0.540	0.275	0.496	-0.130
Logistic Regression	0.435	0.107	0.488	-0.267
Linear SVM	0.339	-0.012	0.463	-0.335
Random Forest	0.661	0.418	0.564	-0.035
Reference Correlation	0.376	0.003	0.499	-0.354

(b) Granularity and error structure

Model	G	R_{sibling}	Cross-lineage
Soft-cascade Transformer	0.174	0.250	0.750
Flat FT-style Transformer	0.176	0.243	0.757
FT-style Transformer + HCE	0.183	0.257	0.743
XGBoost	0.179	0.246	0.754
Logistic Regression	0.167	0.295	0.705
Linear SVM	0.200	0.265	0.735
Random Forest	0.158	0.191	0.809
Reference Correlation	0.099	0.225	0.775

Table 11: Exploratory subtype diagnostics on cell-weighted pooled patient-disjoint predictions. ρ_c and ρ_p are marginal Spearman associations of subtype F1 with log cell count and patient coverage; partial correlations residualize ranked variables. $G = F_1^{\text{coarse}} - F_1^{\text{fine}}$. Among errors, R_{sibling} retains the true parent and Cross-lineage is its complement. These cohort-level summaries are descriptive, not causal.

Table 12 gives all subtype-level quantities and 95% percentile intervals from 2,000 patient-clustered bootstrap resamples. Counts and coverage use pooled ground truth, while F1 uses the same patient-disjoint OOF predictions as Table 2. Mean cells per patient averages only over patients in which the subtype is observed.

Table 12: Per-subtype support and pooled out-of-fold F1 with 95% percentile intervals from 2,000 patient-clustered bootstrap resamples. Cells/patient is the mean among patients in which that subtype is observed. Head and Tail denote the seven most and seven least abundant subtypes; all others are Mid.

Subtype	Lineage	Cells	Patients	Cells/patient	Regime	Soft-cascade F1 [95% CI]	XGB F1 [95% CI]
Erythrocyte	Other	65	6	10.8	Tail	0.253 [0.000, 0.395]	0.133 [0.000, 0.545]
Atypical Memory B	B	781	34	23.0	Tail	0.005 [0.000, 0.014]	0.005 [0.000, 0.014]
CD4 Monocyte-4	Myeloid	1,405	35	40.1	Tail	0.027 [0.006, 0.053]	0.014 [0.000, 0.033]
platelet	Other	1,574	38	41.4	Tail	0.078 [0.016, 0.125]	0.063 [0.015, 0.106]
Tcm	T	2,121	36	58.9	Tail	0.000 [0.000, 0.000]	0.004 [0.000, 0.016]
Memory B-2	B	2,130	33	64.5	Tail	0.017 [0.004, 0.033]	0.011 [0.000, 0.024]
Plasma	B	2,173	40	54.3	Tail	0.611 [0.522, 0.684]	0.599 [0.496, 0.685]
pDC	Myeloid	2,615	40	65.4	Mid	0.029 [0.010, 0.061]	0.019 [0.005, 0.040]
CD14 Monocyte-3	Myeloid	2,977	32	93.0	Mid	0.258 [0.170, 0.306]	0.177 [0.103, 0.211]
CD8 + CTL-3	T	3,244	27	120.1	Mid	0.020 [0.004, 0.035]	0.023 [0.008, 0.043]
GCB	B	3,283	31	105.9	Mid	0.039 [0.007, 0.071]	0.047 [0.001, 0.073]
Cycling T	T	3,667	40	91.7	Mid	0.028 [0.015, 0.043]	0.012 [0.003, 0.022]
CD14 Monocyte-2	Myeloid	3,931	40	98.3	Mid	0.075 [0.046, 0.102]	0.078 [0.056, 0.097]
CD8 + CTL-2	T	4,049	33	122.7	Mid	0.069 [0.017, 0.116]	0.078 [0.015, 0.113]
CD16 Monocyte-2	Myeloid	5,142	40	128.6	Mid	0.111 [0.064, 0.161]	0.117 [0.063, 0.164]
CD8 + CTL-1	T	7,303	40	182.6	Mid	0.009 [0.000, 0.024]	0.002 [0.001, 0.004]
cDC2	Myeloid	7,376	40	184.4	Mid	0.132 [0.099, 0.181]	0.125 [0.079, 0.167]
CD8 Tem-2	T	10,548	40	263.7	Mid	0.019 [0.005, 0.038]	0.017 [0.008, 0.030]
CD16 Monocyte-1	Myeloid	14,678	40	366.9	Mid	0.152 [0.111, 0.198]	0.118 [0.090, 0.150]
CD4 + Tcm-2	T	16,861	39	432.3	Mid	0.188 [0.111, 0.236]	0.188 [0.136, 0.220]
Naive T	T	17,960	40	449.0	Head	0.131 [0.067, 0.183]	0.149 [0.073, 0.199]
Memory B-1	B	23,722	40	593.0	Head	0.034 [0.020, 0.048]	0.032 [0.017, 0.047]
CD8 Tem-1	T	26,714	40	667.9	Head	0.187 [0.128, 0.250]	0.162 [0.107, 0.221]
CD14 Monocyte-1	Myeloid	29,871	40	746.8	Head	0.437 [0.378, 0.495]	0.451 [0.392, 0.507]
CD56 NK	NK	38,442	40	961.0	Head	0.398 [0.320, 0.436]	0.437 [0.373, 0.480]
CD16 NK	NK	58,507	40	1462.7	Head	0.383 [0.326, 0.434]	0.376 [0.317, 0.429]
CD4 + Tcm-1	T	70,653	40	1766.3	Head	0.417 [0.368, 0.460]	0.435 [0.393, 0.473]

For continuity with common long-tail reporting, Table 13 aggregates the seven least and seven most abundant subtypes. This summary is secondary to the per-class audit because individual subtypes can depart sharply from the abundance trend.

Model	Tail Macro-F1	Head Macro-F1
Soft-cascade Transformer	0.142	0.284
Flat FT-style Transformer	0.117	0.290
FT-style Transformer + HCE	0.125	0.288
XGBoost	0.119	0.292
Logistic Regression	0.114	0.197
Linear SVM	0.082	0.204
Random Forest	0.072	0.245
Reference Correlation	0.060	0.114

Table 13: Fine-subtype macro-F1 on the seven least and seven most abundant subtypes under cell-weighted pooled patient-disjoint evaluation. This aggregate is reported as a descriptive complement to the complete per-subtype audit in Table 12.

Cross-lineage error relative to an error-conditioned null. Table 14 uses the wrong-conditioned null as the primary reference: after excluding the true subtype, it renormalizes the predicted-subtype marginal so every randomized outcome remains an error. For the four main models, observed cross-lineage rates are 0.743–0.757 versus wrong-conditioned expectations of 0.741–0.745. The corresponding differences are only +0.002 to +0.013, and all four patient-bootstrap 95% intervals include zero. Thus the raw approximately 75% rate is largely explained by the available wrong-class distribution and does not independently demonstrate either hierarchy preservation or hierarchy failure. The unconditional permutation, which can create chance exact matches and therefore uses a different sample space, is retained only as a sensitivity analysis.

Model	Errors	Observed	Wrong-cond. null	Obs. – WC null [95% CI]	Uncond. null (sens.)
Soft-cascade Transformer	251,354	0.750	0.744	+0.005 [-0.025, +0.028]	0.675
Flat FT-style Transformer	247,798	0.757	0.745	+0.011 [-0.020, +0.035]	0.675
FT-style Transformer + HCE	250,944	0.743	0.741	+0.002 [-0.028, +0.023]	0.671
XGBoost	244,990	0.754	0.741	+0.013 [-0.020, +0.038]	0.672
Logistic Regression	301,566	0.705	0.752	-0.047 [-0.061, -0.036]	0.720
Linear SVM	296,904	0.735	0.774	-0.039 [-0.053, -0.028]	0.738
Random Forest	246,420	0.809	0.749	+0.059 [+0.021, +0.083]	0.685
Reference Correlation	328,596	0.775	0.785	-0.009 [-0.022, +0.003]	0.758

Table 14: Cross-lineage error relative to an error-conditioned marginal null. The primary wrong-conditioned (WC) null draws from each model’s predicted-subtype marginal after excluding the true subtype and renormalizing, so every randomized outcome remains wrong. Observed – WC-null confidence intervals recompute both terms in 2,000 patient-clustered bootstrap resamples. The unconditional permutation preserves the realized true- and predicted-subtype marginals but can create chance exact matches; it is reported only as a sensitivity analysis.

C.8 PROTOCOL SENSITIVITY: PATIENT-LEVEL VERSUS CELL-LEVEL SPLITTING

Table 1 compares the primary patient-disjoint protocol with a leakage-prone cell-level split control. It is placed in the appendix because the central result uses patient-disjoint evaluation throughout; the control quantifies how much the easier protocol would inflate both absolute performance and the apparent advantage of a model family.

Metric definitions. Let $C^{(p)} \in \mathbb{R}^{K \times K}$ be the confusion matrix for held-out subject p and $n_p = \sum_{ij} C_{ij}^{(p)}$. We distinguish

$$C^{\text{CW}} = \sum_p C^{(p)}, \quad F_1^{\text{CW}} = \text{MacroF1}(C^{\text{CW}}), \quad (12)$$

$$C^{\text{SB}} = \frac{1}{P} \sum_p \frac{C^{(p)}}{n_p}, \quad F_1^{\text{SB}} = \text{MacroF1}(C^{\text{SB}}), \quad (13)$$

$$F_1^{\text{within}} = \frac{1}{P} \sum_p \text{MacroF1}(C^{(p)}). \quad (14)$$

We call these *cell-weighted pooled Macro-F1*, *subject-balanced pooled Macro-F1*, and *mean within-subject Macro-F1*. Subject balancing normalizes each subject confusion matrix to unit mass before pooling; it is neither equal-cell subsampling nor averaging per-subject Macro-F1. Released files retain the legacy column name `patient_balanced_macro_f1` for F_1^{SB} , but no different patient-balanced statistic is used.

The numerical split comparison appears in the main results (Table 1). Its historical scores are cell-weighted in both columns, rather than the subject-balanced leaderboard metric.

Equal-cell evaluation sensitivity. Patient-disjoint evaluation prevents leakage but does not make a cell-weighted pooled point estimate subject-balanced; patient cell counts range from 1,929 to 22,627. Table 15 therefore recomputes both hierarchy endpoints after sampling 1,900 cells without replacement from every patient for each of 50 seeds. Coarse macro-F1 changes by at most +0.005, while fine macro-F1 changes by −0.007 to −0.009; the qualitative ordering in this sensitivity is unchanged. These small changes describe robustness of the historical cell-weighted scores to equal-cell sampling; they do not prove absence of patient influence. The main table now uses exact subject-normalized confusion pooling rather than this Monte Carlo subsampling approximation.

Model	Input	Coarse SB-Macro-F1 $\uparrow$	Fine SB-Macro-F1 $\uparrow$
Soft-cascade Transformer	APT	0.332 [0.306, 0.354]	0.145 [0.122, 0.160]
Flat Transformer	APT	0.326 [0.300, 0.348]	0.139 [0.114, 0.155]
HCE Transformer	APT	0.335 [0.310, 0.356]	0.143 [0.121, 0.160]
XGBoost	APT	0.325 [0.302, 0.345]	0.137 [0.120, 0.149]
Linear SVM	APT	0.309 [0.292, 0.324]	0.105 [0.089, 0.118]
Logistic Regression	APT	0.298 [0.281, 0.315]	0.126 [0.107, 0.139]
Random Forest	APT	0.253 [0.227, 0.277]	0.089 [0.079, 0.097]
Reference Correlation	APT	0.171 [0.147, 0.192]	0.068 [0.050, 0.078]

Table 16: Complete subject-balanced pooled Macro-F1 reference inventory, including auxiliary classical baselines. Definitions, resampling, and scope match Table 2.

Model	Input	Coarse CW-Macro-F1 $\uparrow$	Fine CW-Macro-F1 $\uparrow$
Soft-cascade Transformer	APT	0.326 [0.300, 0.350]	0.152 [0.129, 0.166]
Flat Transformer	APT	0.323 [0.299, 0.344]	0.147 [0.120, 0.161]
HCE Transformer	APT	0.333 [0.310, 0.354]	0.151 [0.127, 0.166]
XGBoost	APT	0.322 [0.298, 0.343]	0.143 [0.127, 0.158]
Linear SVM	APT	0.311 [0.296, 0.325]	0.111 [0.093, 0.126]
Logistic Regression	APT	0.299 [0.283, 0.315]	0.132 [0.113, 0.146]
Random Forest	APT	0.250 [0.228, 0.271]	0.092 [0.082, 0.100]
Reference Correlation	APT	0.172 [0.149, 0.189]	0.072 [0.055, 0.082]

Table 17: Historical cell-weighted pooled Macro-F1, with pointwise 95% patient-bootstrap intervals, for the frozen APT-only hierarchy predictions. The same predictions are reweighted in Table 2; the estimands differ. Classical auxiliary references are retained here rather than omitted from the benchmark.

Task	Model	CW-pooled M-F1	Equal-cell M-F1 [95% interval]	Δ
Coarse lineage	Soft-cascade Transformer	0.326	0.331 [0.325, 0.337]	+0.005
	Flat FT-style Transformer	0.323	0.326 [0.322, 0.330]	+0.003
	FT-style Transformer + HCE	0.333	0.334 [0.328, 0.340]	+0.000
	XGBoost	0.322	0.324 [0.320, 0.329]	+0.002
Fine subtype	Soft-cascade Transformer	0.152	0.144 [0.138, 0.153]	-0.008
	Flat FT-style Transformer	0.147	0.138 [0.131, 0.147]	-0.008
	FT-style Transformer + HCE	0.151	0.142 [0.137, 0.151]	-0.009
	XGBoost	0.143	0.136 [0.131, 0.144]	-0.007

Table 15: Equal-cell evaluation sensitivity. For each of 50 seeds, 1,900 cells are sampled without replacement from every subject, and cell-weighted pooled Macro-F1 is recomputed on the resulting 76,000 cells. All models use identical sampled row indices. The interval is the empirical 2.5th–97.5th percentile across seeds; Δ is the equal-cell mean minus the original cell-weighted pooled metric. This resampling diagnostic is not the subject-balanced pooled metric, and it does not retrain or reselect models.

C.9 COARSE–FINE CONSISTENCY DIAGNOSTIC

Independent coarse and fine classifiers can disagree about the hierarchy even when each prediction is plausible in isolation. Table 18 reports their natural agreement on the same frozen OOF predictions as the main results. These auxiliary path metrics remain cell-weighted averages, whereas the main Macro-F1 table now balances patient contributions.

The main reporting metric was revised retrospectively without retraining, retuning, or reselecting checkpoints. The historical cell-weighted comparison family contrasted Soft-cascade with HCE and XGBoost at coarse and fine resolution, using 2,000 paired patient bootstrap resamples and Holm correction over four tests; none reached adjusted significance. Those tests do not apply to the subject-balanced results. Flat and auxiliary classical comparisons remain descriptive. The saved

re-evaluation script reproduces the old point estimates before generating the new table, and records source-artifact hashes.

Model	Fine Acc.	Root-excluded Hier. F1	Consistency	Exact Path	Subtype Tree Dist.
Soft-cascade Transformer	0.305	0.404	0.812	0.253	2.431
Flat FT-style Transformer	0.315	0.411	0.795	0.257	2.407
FT-style Transformer + HCE	0.306	0.405	0.834	0.259	2.418
XGBoost	0.323	0.420	0.736	0.248	2.375

Table 18: Cell-weighted auxiliary hierarchy diagnostics on the same frozen OOF predictions as Table 2. Root-excluded hierarchical F1 is the example-averaged F1 between true and predicted {lineage, subtype} sets; the always-correct root is omitted. Exact Path requires both independent predictions to be correct. Consistency compares the mapped fine parent with the independent coarse prediction. Subtype Tree Distance uses only the predicted subtype and its mapped parent: 0/2/4 for correct/sibling/cross-lineage predictions (lower is better). *Unknown* is excluded; no constrained decoding is used.

C.10 ARCHIVED FEATURE-REDUNDANCY DIAGNOSTIC

We retain this audit only to document how easily the unresolved within-cohort signal can be compressed. Development-selected ten-feature panels reproduced the observed full-panel result on the same 40 patients, but random panels also performed strongly. This is neither biomarker discovery nor evidence that a reduced assay or low-cell clinical workflow is valid. Feature overlap across folds cannot distinguish a reproducible biological signature from a shared acquisition signature.

The complete historical protocol, curve, and random-panel, TreeSHAP, and cell-subsampling tables are preserved in the repository at `archive/compact_panel_audit.tex`; they are omitted from this PDF to avoid giving a secondary diagnostic the appearance of an assay-efficiency contribution. The archive records the cell-weighted LR ranker, conditional inner budget selection, foldwise refitting, and random-panel construction. Feature restriction involved refitting; inference-cell caps instead sampled cached held-out predictions after training on all development cells. Neither feature stability nor a 100-cell aggregation result establishes molecular specificity, training-data efficiency, or clinical validity. Such claims would require acquisition metadata and independent APT validation with a frozen pipeline.

C.11 BALANCED SAMPLING ABLATION

A development-only class-balanced-sampling variant did not outperform natural sampling after matching optimization steps and was not retained.

C.12 EXPLORATORY LEAVE-ONE-DISEASE-OUT STRESS TEST

We retain only a leave-one-disease-out (LODO) representation stress test. For each run, all patients from one disease category are withheld from training and used only for evaluation. Table 19 shows that generalization weakens substantially relative to the standard patient-disjoint folds, especially for fine subtypes. Because there are only six disease groups and no external cohort, LODO is descriptive rather than an estimate of deployment robustness.

Held-out Disease	Fine Macro-F1	Fine Acc.	Coarse Macro-F1	Coarse Acc.	Tail Macro-F1
BTC	0.086	0.317	0.310	0.479	0.073
CRC	0.105	0.277	0.309	0.520	0.112
GC	0.063	0.269	0.281	0.488	0.057
HCC	0.063	0.246	0.264	0.383	0.072
PC	0.057	0.253	0.221	0.482	0.037
Macro-average	0.075	0.272	0.277	0.470	0.070

Table 19: Leave-one-disease-out generalization for Soft-cascade Transformer over the five malignant held-out diseases. Each row withholds all patients of the named disease from training. The in-distribution reference using the same model under the standard patient-disjoint 5-fold protocol is fine Macro-F1 0.152 and coarse Macro-F1 0.326. A separate Normal-held-out run produces near-chance fine performance. LODO is reported only as an exploratory appendix stress test, not as proof that the model already solves cross-disease generalization.

We omit unmatched detector rankings. Cross-disease generalization remains unresolved and requires external cohorts with prespecified distribution shifts and patient-clustered uncertainty.